INCH-POUND

MS9090E
10 OCTOBER 2021
SUPERSEDING
MS9090D
4 AUGUST 2011

DETAIL SPECIFICATION SHEET

BOLT MACHINE, DRILLED 12 POINT HEAD, STEEL, .3125-24 UNF-3A

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and procurement specification SAE-AS7452.

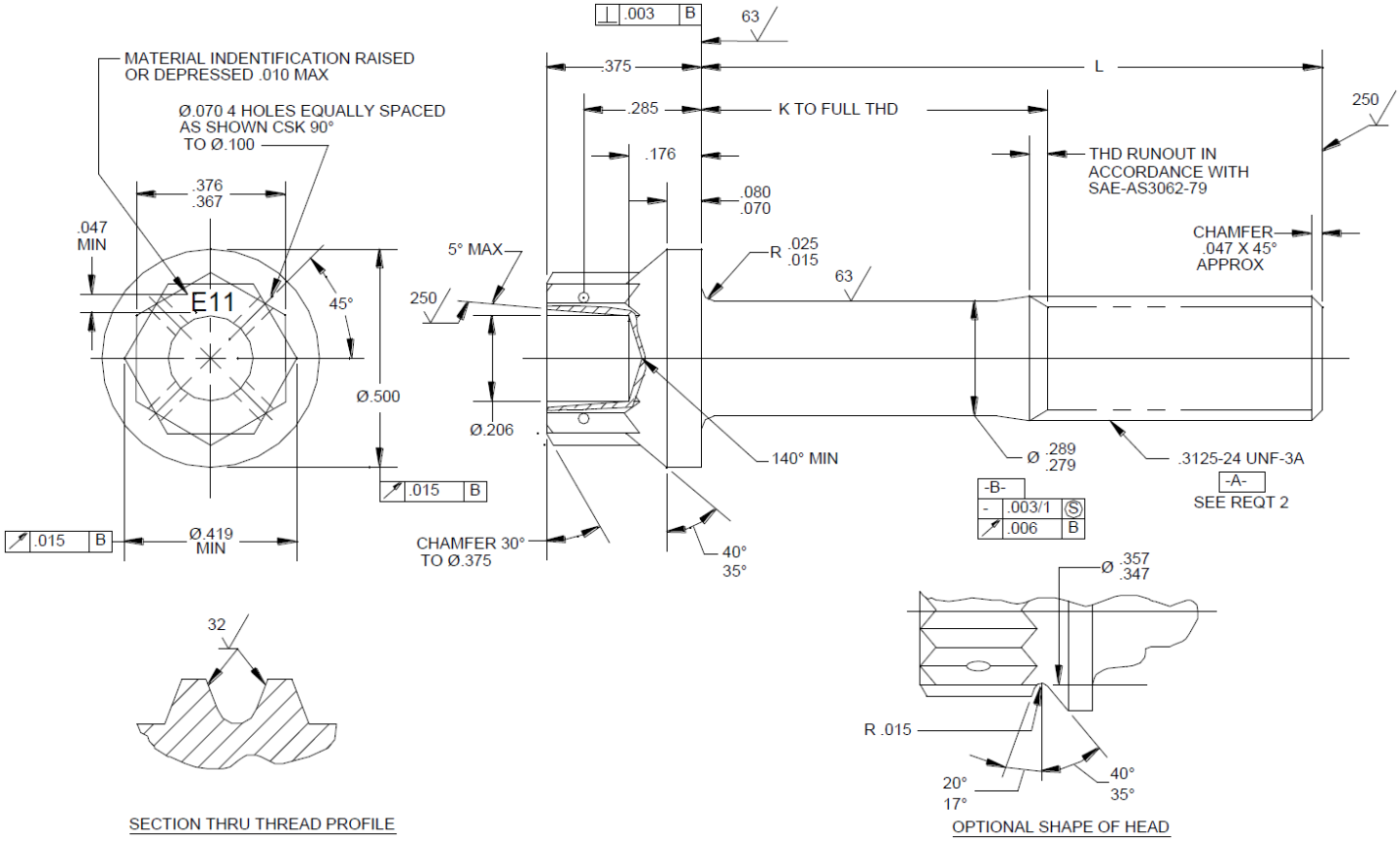


FIGURE I. BOLT

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AMSC N/A FSC 5306

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TABLE I. Dash numbers and dimensions

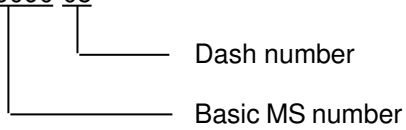
DASH NUMBER	L	K +.000 -.060	APPROX WEIGHT LBS/100
-05	.562	1/	2.21
-06	.625	1/	2.28
-07	.688	1/	2.39
-08	.750	1/	2.51
-09	.812	1/	2.62
-10	.875	1/	2.73
-11	.938	1/	2.85
-12	1.000	.125	2.96
-13	1.062	.188	3.07
-14	1.125	.250	3.19
-15	1.188	.312	3.30
-16	1.250	.375	3.41
-17	1.312	.438	3.52
-18	1.375	.500	3.64
-19	1.438	.562	3.75
-20	1.500	.625	3.87
-21	1.562	.688	3.98
-22	1.625	.750	4.10
-23	1.688	.812	4.22
-24	1.750	.875	4.33
-25	1.875	1.000	4.55
-26	2.000	1.125	4.77
-27	2.125	1.250	5.00
-28	2.250	1.375	5.23
-29	2.375	1.500	5.45
-30	2.500	1.625	5.67
-31	2.625	1.750	5.90
-32	2.750	1.875	6.13
-33	2.875	2.000	6.36
-34	3.000	2.125	6.58
-35	3.125	2.250	6.80
-36	3.250	2.375	7.03
-37	3.375	2.500	7.27
-38	3.500	2.625	7.50
-39	3.625	2.750	7.73
-40	3.750	2.875	7.95
-41	3.875	3.000	8.17
-42	4.000	3.125	8.40
-43	4.125	3.250	8.63
-44	4.250	3.375	8.86
-45	4.375	3.500	9.08
-46	4.500	3.625	9.30
-47	4.625	3.750	9.52
-48	4.750	3.875	9.74
-49	4.875	4.000	9.95
-50	5.000	4.125	10.17
-51	5.125	4.250	10.40
-52	5.250	4.375	10.63
-53	5.375	4.500	10.86
-54	5.500	4.625	11.10
-55	5.625	4.750	11.33
-56	5.750	4.875	11.56
-57	5.875	5.000	11.78
-58	6.000	5.125	12.00

1/ Thread to head. Maximum two incomplete threads.

Requirements:

1. Material: Steel in accordance with SAE-AMS 6322-80.
2. Threads: Threads in accordance with MIL-S-7742. Incomplete threads not to enter fillet.
3. Finish: Cadmium plate in accordance with SAE-AMS 2400-80. Dimensions specified are after plating.
4. Hardness: 26-32 HRC.
5. Dimensioning and tolerancing: Dimensioning and tolerancing shall be in accordance with ASME Y14.5M.
6. Wrenching configuration: Wrenching configuration in accordance with SAE-AS870-74.
7. Surface texture: Surface texture, unless otherwise specified, shall be 125 microinches, except upset head, in accordance with ASME B46.1.
8. Magnetic particle inspection: Magnetic particle inspection shall be in accordance ASTM-E1444.
9. Tolerance: Unless otherwise specified, tolerances shall be for linear dimensions ±.010, angular dimensions ±5°.
10. Edges: Break sharp edges .003 - .015, unless otherwise specified.
11. Part number: The part number shall consist of the basic MS number followed by a dash number from Table I.

Example: MS9090-05



MS9090-05 indicates: Bolt, Machine, Steel Drilled 12 Point Head; .3125-24 UNF-3A; Length .562.

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Notes:

1. All dimensions are in inches.

2. Do not use unassigned part numbers.

3. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.

4. Referenced Government (or nongovernment) documents of the issue listed in that issue of the department of defense index of specifications and standards (DoDISS) specified in the solicitation form a part of this standard to the extent specified herein.

5. This standard was developed cooperatively with the SAE aerospace propulsion division.

6. AS and AMS are SAE publications.

7. Subject term (key word) listing.

   Cadmium Plate

8. Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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APPENDIX A
3D Model for reference only.

Viewing this model requires a PDF viewer that supports 3D models.
Adobe Acrobat is known to support 3D models but there may be other suitable viewers.

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